
NYCU IMF - Financial Econometrics

Name: Chan Nok Hang
Student Number: 414707007

Due Date: March 10, 2026
Assignment: 1, Ch2 q1, 22, 25

Question 1

Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4				
2	2				
1	3				
-1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum (x_i - \bar{x}) =$	$\sum (x_i - \bar{x})^2 =$	$\sum (y_i - \bar{y}) =$	$\sum (x - \bar{x})(y - \bar{y}) =$

- (a) Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

The sample means $\bar{x} = 1$, and $\bar{y} = 2$.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x - \bar{x})(y - \bar{y}) = 8$

- (b) Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

$$b_2 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x_i - \bar{x})^2} \quad (2.7)$$

$$b_1 = \bar{y} - b_2 \bar{x} \quad (2.8)$$

From part a, using (2.7) and (2.8),

$$b_2 = 8/10 = 0.8$$

$$b_1 = 2 - 0.8 \times 1 = 1.2$$

- (c) Compute $\sum_{i=1}^5 x_i^2$, $\sum_{i=1}^5 x_i y_i$. Using these numerical values, show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum (x - \bar{x})(y - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

$$\sum_{i=1}^5 x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15$$

$$\sum_{i=1}^5 x_i y_i = 3 \times 4 + 2 \times 2 + 1 \times 3 + (-1) \times 1 + 0 \times 0 = 18$$

From part a, $\sum (x_i - \bar{x})^2 = 10$, then
 $\because \sum x_i^2 - N\bar{x}^2 = 15 - 5(1)^2 = 10$,
 $\therefore \sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$

From part a, $\sum (x - \bar{x})(y - \bar{y}) = 8$, then
 $\because \sum x_i y_i - N\bar{x}\bar{y} = 18 - 5 \times 1 \times 2 = 8$
 $\therefore \sum (x - \bar{x})(y - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$

- (d) Use the least squares estimates from part (b) to compute the fitted values of y , and complete the remainder of the table below. Put the sums in the last row.
 Calculate the sample variance of y , $S_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1)$, the sample variance of x , $S_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, the sample covariance between x and y , $S_{xy} = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) / (N - 1)$, the sample correlation between x and y , $r_{xy} = S_{xy} / (S_x S_y)$ and the coefficient of variation of x , $CV_x = 100(S_x / \bar{x})$. What is the median, 50th percentile, of x ?

x	y	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4				
2	2				
1	3				
-1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum \hat{y}_i =$	$\sum \hat{e}_i =$	$\sum \hat{e}_i^2 =$	$\sum x_i \hat{e}_i =$

x	y	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

From part b, $\hat{y}_i = 1.2 + 0.8x_i$, then
 $S_y^2 = \{(4 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 + (1 - 2)^2 + (0 - 2)^2\} / (5 - 1) = 2.5$

$$\begin{aligned}
S_x^2 &= 10/(5-1) = 2.5 \\
S_{xy} &= 8/(5-1) = 2 \\
r_{xy} &= 2/\sqrt{2.5}\sqrt{2.5} = 0.8 \\
CV_x &= 100(\sqrt{2.5}/1) = 158.1139 \\
\text{Median of } x &= 1
\end{aligned}$$

- (e) On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2x_i$.

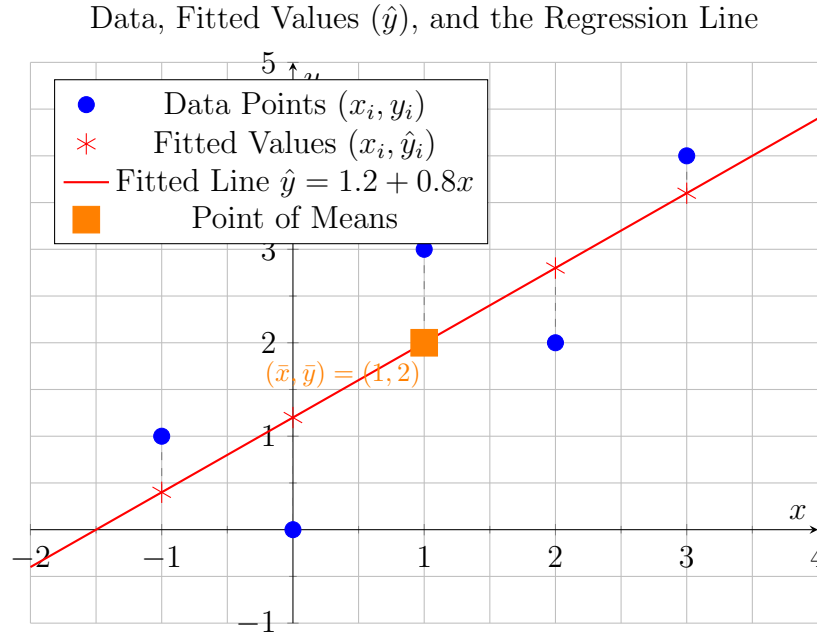


Figure 1: OLS Regression Results showing residuals as dashed lines.

- (f) On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.

As shown in Figure 1, the point of the means $(\bar{x}, \bar{y}) = (1, 2)$ (highlighted with the orange square) lies exactly on the fitted red regression line. This is an inherent property of the Ordinary Least Squares (OLS) estimator.

- (g) Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.

$$\begin{aligned}
&\because \bar{y} = 2, \\
&b_1 + b_2\bar{x} = 1.2 + 0.8(1) = 2 \\
&\therefore \bar{y} = b_1 + b_2\bar{x}
\end{aligned}$$

- (h) Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i/N$.

$$\begin{aligned}
&\because \bar{y} = 2, \\
&\bar{\hat{y}} = (3.6 + 2.8 + 2 + 0.4 + 1.2)/5 = 2
\end{aligned}$$

$$\therefore \hat{y} = \bar{y}$$

- (i) Compute $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{\sum e_i^2}{N-2} = \frac{3.6}{5-2} = 1.2$$

- (j) Compute $\text{var}(b_2|x)$ and $\text{se}(b_2)$.

$$\text{var}(b_2|x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$$

$$\text{se}(b_2) = \sqrt{0.12} \approx 0.3464$$

Question 22

A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star5_small*. [Note: The data file *star5* contains more observations and variables.]

- (a) Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $TOTALSCORE_i = \beta_1 + \beta_2 SMALL + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?

Table 1: OLS Regression Results: Effect of Small Class Size on Total Score

Variable	Coefficient	Std. Error	t-stat	p-value
Intercept (β_1)	916.442	3.675	249.40	< 0.001
Small Class (β_2)	12.175	5.369	2.268	0.0236
R^2	0.0066			
Observations	775			

The regression analysis indicates that small class sizes have a positive and statistically significant impact on student achievement ($p < 0.05$). Specifically, students assigned to small classes scored, on average, approximately 12.18 points higher on their combined aptitude tests compared to their peers in regular-sized classes.

- (b) Repeat part (a) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

Table 2: OLS Regression Results: Effect of Small Class Size by Subject

Predictor	Dependent Variable	
	Reading Score	Math Score
Intercept (β_1)	432.665*** (1.488)	483.777*** (2.449)
Small Class (β_2)	6.924** (2.174)	5.251 (3.578)
Observations	775	775
R^2	0.0130	0.0028

Standard errors in parentheses.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Yes, there is a notable difference. The positive effect of a small class size on the totalscore observed in part (a) is heavily driven by improvements in reading. While small classes produce a statistically significant increase of ~ 6.9 points in reading ($p < 0.01$), the estimated increase of ~ 5.3 points in math is not statistically distinguishable from zero ($p = 0.143$).

- (c) Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?

Table 3: OLS Regression Results: Effect of a Teacher Aide on Total Score

Variable	Coefficient	Std. Error	t -stat	p -value
Intercept (γ_1)	916.442	3.559	257.53	< 0.001
Teacher Aide (γ_2)	4.306	4.994	0.862	0.389
R^2	0.0009			
Observations	837			

Based on the regression results, adding a full-time teacher's aide to a regular-sized classroom has no statistically significant effect on students' overall learning ($p = 0.389$). While the estimated coefficient is slightly positive (4.306 points), we cannot confidently distinguish this effect from zero. Therefore, unlike reducing the actual class size, merely adding an aide does not appear to be an effective intervention for improving combined aptitude scores in this cohort.

- (d) Repeat part (c) using dependent variables READSCORE and MATHSCORE. Do you observe any differences?

Table 4: OLS Regression Results: Effect of a Teacher Aide by Subject

Predictor	Dependent Variable	
	Reading Score	Math Score
Intercept (γ_1)	432.665*** (1.460)	483.777*** (2.355)
Teacher Aide (γ_2)	2.871 (2.049)	1.435 (3.304)
Observations	837	837
R^2	0.0023	0.0002

Standard errors in parentheses.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

Yes, a clear difference emerges when comparing the interventions. In part (b), assigning students to a small class yielded a statistically significant improvement in reading scores ($p < 0.01$). In contrast, assigning a full-time teacher's aide to a regular-sized class fails to produce a statistically significant improvement in either reading ($p = 0.161$) or math ($p = 0.664$). The estimated coefficients for the aide are small and statistically indistinguishable from zero across both subjects, indicating that reducing class size is a much more effective educational intervention than adding a teacher's aide.

Question 25

Consumer expenditure data from 2013 are contained in the file `cex5_small`. [Note: `cex5` is a larger version with more observations and variables.] Data are on three-person households consisting of a husband and wife, plus one other member, with incomes between 1000 *per month* to 20,000 per month. *FOODAWAY* is past quarter's food away from home expenditure per month per person, in dollars, and *INCOME* is household monthly income during past year, in \$100 units.

- (a) Construct a histogram of *FOODAWAY* and its summary statistics. What are the mean and median values? What are the 25th and 75th percentiles?

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	12.04	32.55	49.27	67.5	1179

Table 5: Question 25 part a Summary statistics

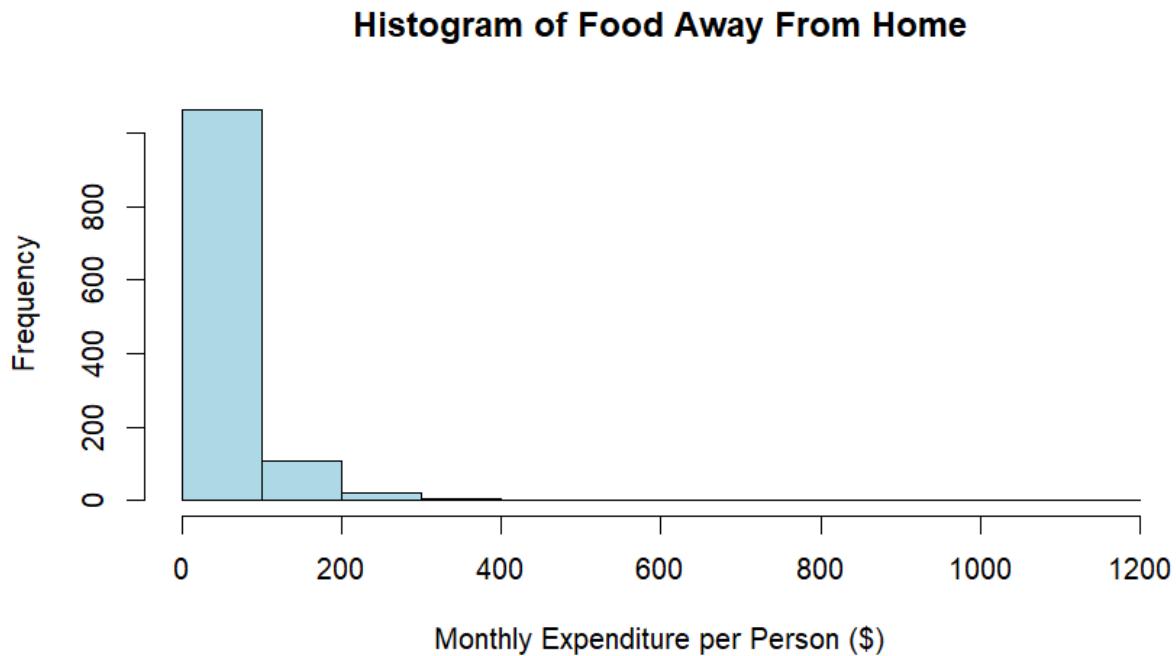


Figure 2: Question 25 part a: Histogram of *FOODAWAY*

According to the summary statistic, the mean is 49.27, and the median is 32.5550. The 25th percentile is 12.0400, and the 75th percentile is 67.5025.

- (b) What are the mean and median values of *FOODAWAY* for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?

For households including a member with an advanced degree, the *FOODAWAY* has:

- Mean: 73.15494
- Median: 48.15

For households including a member with a college degree, the *FOODAWAY* has:

- Mean: 48.59718
- Median: 36.11

For households including a member with no advanced or college degree, the *FOODAWAY* has:

- Mean: 39.01017
- Median: 26.02

- (c) Construct a histogram of $\ln(\text{FOODAWAY})$ and its summary statistics. Explain why FOODAWAY and $\ln(\text{FOODAWAY})$ have different numbers of observations.

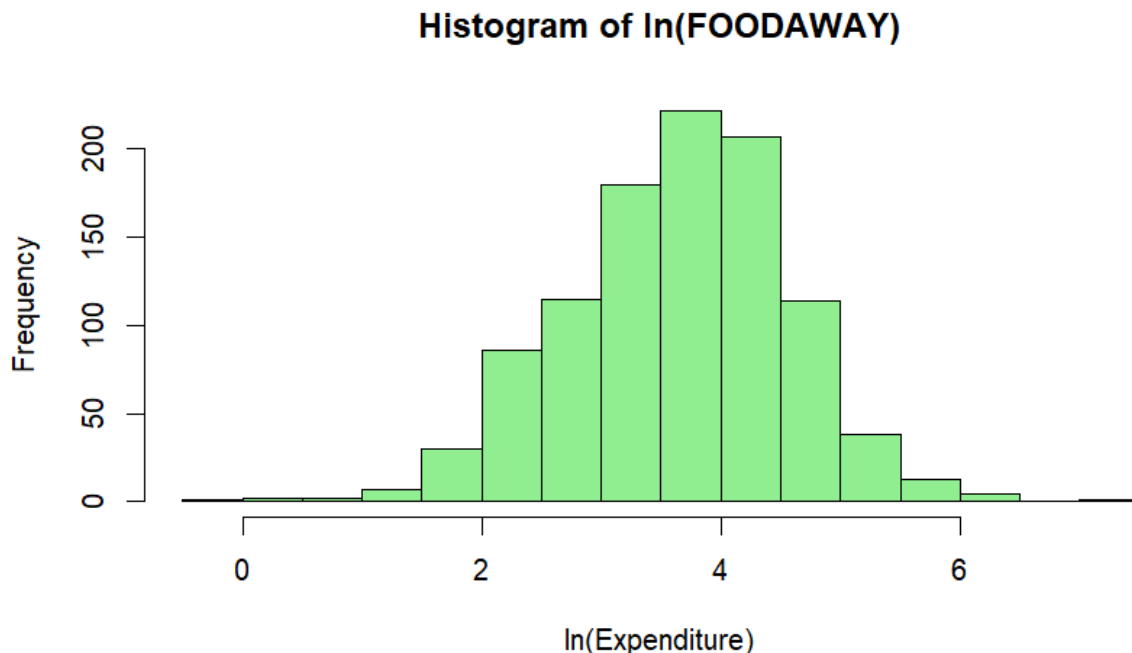


Figure 3: Question 25 part c: Histogram of $\ln(\text{FOODAWAY})$

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-0.3011	3.0759	3.6865	3.6508	4.2797	7.0724

Table 6: Question 25 part c Summary statistics

The variables FOODAWAY and $\ln(\text{FOODAWAY})$ have different numbers of observations because the natural logarithm of zero is mathematically undefined. In this dataset, some households reported spending exactly \$0 on food away from home during the quarter. When applying the log transformation, these zero-expenditure observations either result in negative infinity ($-\text{Inf}$ in R) or must be dropped from the dataset entirely to compute summary statistics and run a log-linear regression.

- (d) Estimate the linear regression $\ln(\text{FOODAWAY}) = \beta_1 + \beta_2 \text{INCOME} + e$. Interpret the estimated slope.

Based on the regression result, the estimated regression model is:

$$\widehat{\ln(\text{FOODAWAY})} = 3.1293 + 0.0069017 \times \text{INCOME}.$$

Table 7: OLS Regression Results: Log-Linear Model of Food Expenditure

Variable	Coefficient	Std. Error	<i>t</i> -stat	<i>p</i> -value
Intercept (β_1)	3.1293	0.0566	55.34	< 0.001
INCOME (β_2)	0.0069	0.0007	10.54	< 0.001
R^2	0.0983			
Observations	1022			

For every additional unit of *INCOME* (which corresponds to a \$100 increase in actual monthly household income), the monthly per-person expenditure on food away from home (*FOODAWAY*) is estimated to increase by approximately 0.69% (0.0069017×100), holding all else constant. This effect is highly statistically significant ($p < 0.001$).

- (e) Plot $\ln(\text{FOODAWAY})$ against *INCOME*, and include the fitted line from part (d).

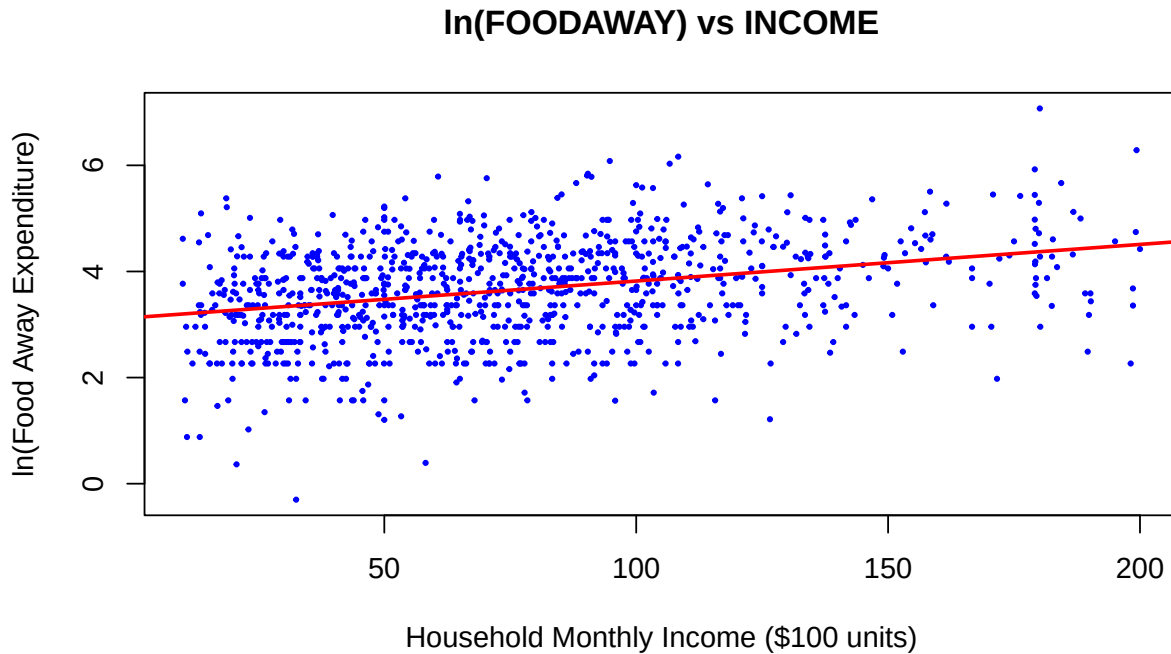


Figure 4: Question 25 part e

- (f) Calculate the least squares residuals from the estimation in part (d). Plot them vs. *INCOME*. Do you find any unusual patterns, or do they seem completely random?

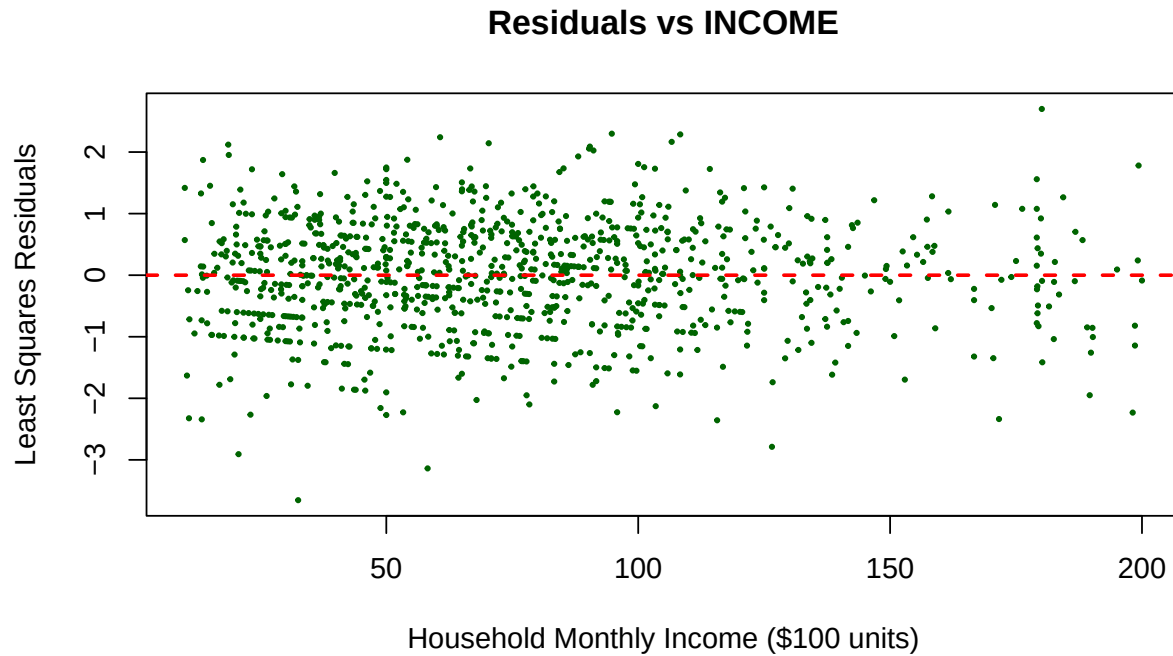


Figure 5: Question 25 part f

An examination of the plot of the least squares residuals against *INCOME* reveals that the points appear randomly scattered around the zero-mean line. There is no distinct 'fan' shape (which would indicate heteroskedasticity) or curved pattern (which would indicate non-linearity). The vertical spread of the residuals remains relatively constant across all income levels. This suggests that applying the natural logarithm to the *FOODAWAY* expenditure successfully stabilized the variance, resulting in a largely homoskedastic error term that satisfies the classical assumptions of OLS.